Incident Handler's Journal

Entry #1: Documenting a Cybersecurity Incident

Date: October 19, 2026

Entry: Documenting a cybersecurity incident

Description:

A healthcare company experienced a ransomware attack after an employee clicked a suspicious link in a phishing email. The incident affected the company's computer systems and required the incident response team to investigate and contain the threat. The team worked to isolate affected systems and restore normal operations.

The 5 W's

- **Who:** The healthcare company, its employees, and the incident response team.
- **What:** A ransomware attack affected the company's computer systems.
- **Where:** The healthcare company's network.
- **When:** After an employee clicked a suspicious phishing link.
- **Why:** The phishing email provided a way for the attacker to compromise the systems.

Additional notes:

This activity helped me understand why documenting an incident is important. Accurate records help security teams understand what happened, identify the cause, and decide what actions are needed to prevent similar incidents.

Entry #2: Analyzing a Packet Capture File

Date: October 19, 2026

Entry: Analyzing a packet capture file

Description:

I used Wireshark to examine a packet capture file and learn how network traffic can be analyzed. I explored packets and their details to understand how devices communicate over a network.

The 5 W's

Not applicable. This activity focused on learning a cybersecurity tool rather than investigating a specific incident.

Additional notes:

Using Wireshark helped me understand how network protocol analyzers can help security analysts inspect network traffic. At first, the amount of information was difficult to understand, but examining packet details helped me become more familiar with the tool.

Entry #3: Capturing My First Packet

Date: October 20, 2026

Entry: Capturing my first packet

Description:

I used tcpdump to capture network packets through the command line. I practiced selecting a network interface, filtering traffic, limiting the number of packets captured, and saving the capture to a file.

The 5 W's

Not applicable. This activity focused on practicing packet capture commands rather than investigating a specific incident.

Additional notes:

This activity helped me learn how tcpdump works and how command options affect the results. I also learned that small mistakes in command syntax can change the output, so it is important to read commands carefully and understand each option.

Entry #4: Investigating a Suspicious File Hash

Date: October 20, 2026

Entry: Investigating a suspicious file hash

Description:

I used VirusTotal to investigate a suspicious file hash in a scenario involving a financial services company. I reviewed the available detection results to understand whether the file was associated with malicious activity and how threat intelligence can support an investigation.

The 5 W's

- **Who:** The financial services company and its security team.
- **What:** A suspicious file was investigated using its SHA-256 hash.
- **Where:** The investigation concerned a file associated with the company's environment.
- **When:** During the security investigation.
- **Why:** The file needed to be checked for possible malicious activity.

File hash:

```
54e6ea47eb04634d3e87fd7787e2136ccfbcc80ade34f246a12cf93bab527f6b
```

Additional notes:

This activity helped me understand how file hashes can be used as indicators of compromise. I learned that threat intelligence tools such as VirusTotal can help analysts review detection information and support incident investigation. I also learned why employees should be cautious when opening unexpected email attachments or links.

Reflections and Notes

During this week, I worked on different cybersecurity activities, including analyzing network traffic with Wireshark, capturing packets using tcpdump, and investigating a suspicious file hash with VirusTotal. These activities gave me practical experience with cybersecurity tools and helped me understand how security analysts investigate suspicious activity and document their findings.

1. Challenging Part

The most challenging part for me was using command-line tools such as tcpdump because I had to understand the commands, their options, and how they affect the results. I overcame this by carefully following the instructions, checking my commands, and repeating the steps when needed. With practice, I became more comfortable using the terminal.

2. Understanding of Incident Response

Before completing these activities, I had a basic understanding of incident response. Through this work, I learned how security analysts identify suspicious activity, investigate evidence, and

document incidents. I also learned about the different phases of incident response, including detection and analysis, containment, eradication, recovery, and lessons learned.

3. Activity I Enjoyed the Most

I enjoyed working with Wireshark and VirusTotal because I could explore network packets and investigate a suspicious file hash. These activities helped me understand how cybersecurity tools are used to analyze potential threats. I found the practical work interesting and would like to continue practicing these tools to improve my investigation skills.